Industrialization and Cancer in Louisiana

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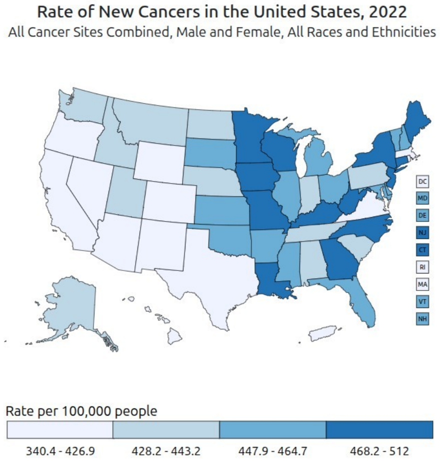
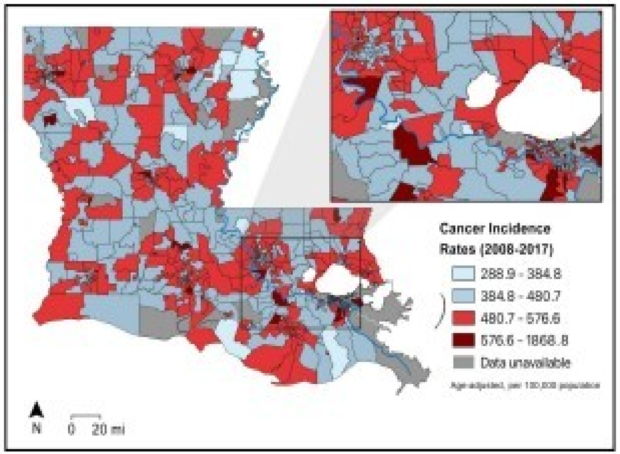
CMPS 3160: Intro to Data Science

Motivations

- "***Air pollution is linked to higher cancer rates among black or impoverished communities in Louisiana***" **by Kimberly A Terrell and Gianna St Julien**
- **Are these claims true? Can we visualize these claims and uphold them?**
- **Our analysis contributes to the growing body of evidence that Black and Brown communities in Louisiana are overburdened with the negative effects toxic air pollution from petrochemical facilities and other sources...**

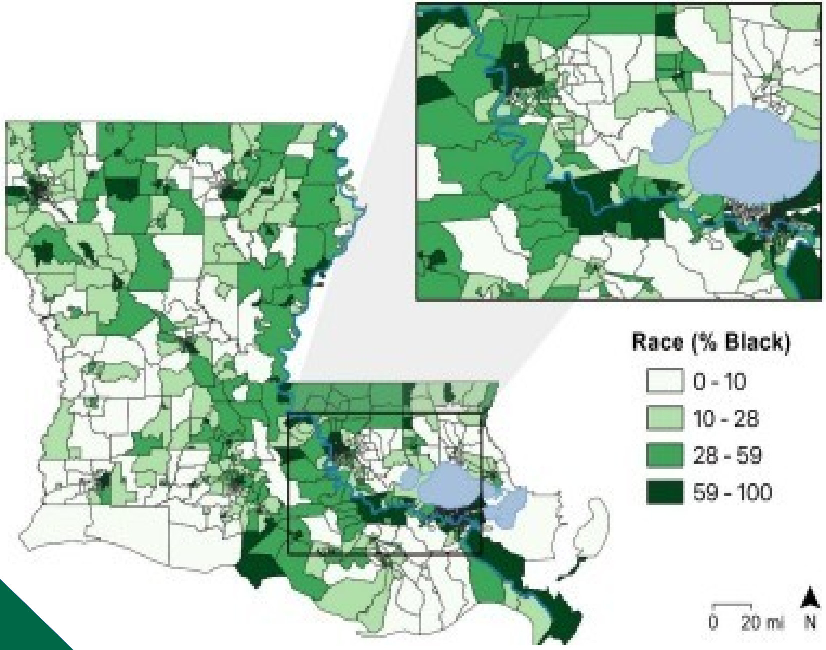


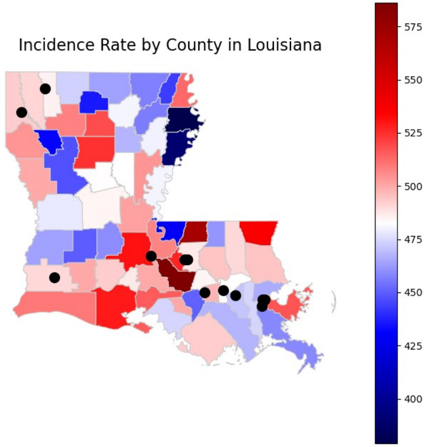


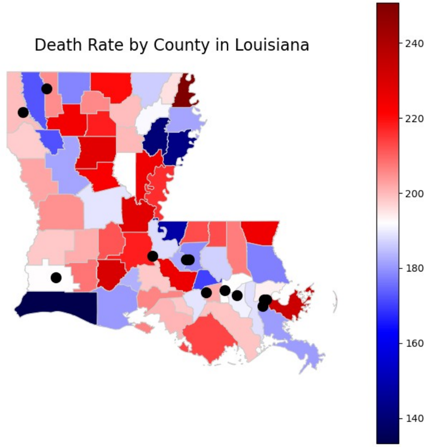
Data Collection and ETL

- Census Bureau’s Community Survey
- Clinical Trials and Cancer Studies conducted by cancer.gov
- Accumulated for analysis on data.world

EDA – On The Right Track



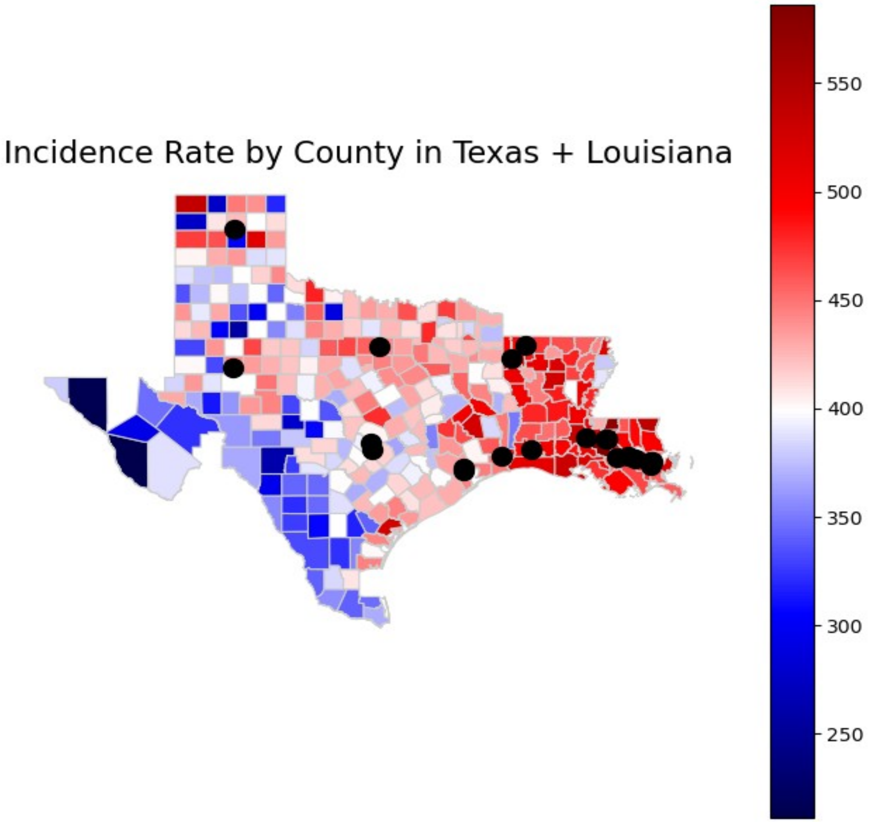


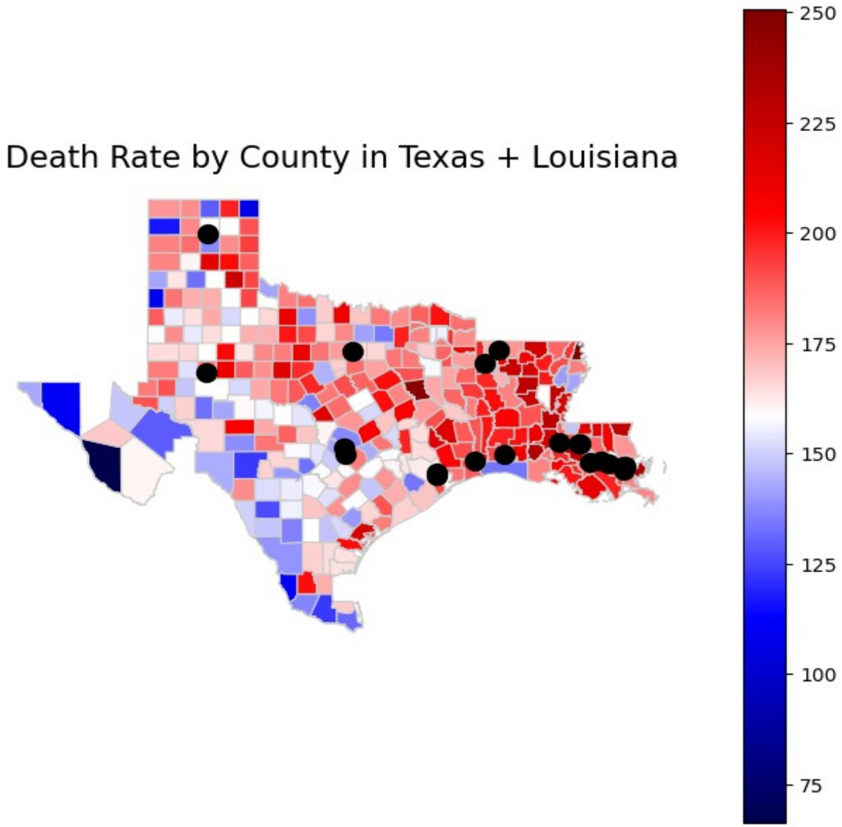



```python
#points of petrochemical refineries - https://www.industryselect.com/blog/top-10-manufacturing-companies-in-louisiana
#it would be great if there was a dataset of these somewhere but I have been unable to find any
points_df = pd.DataFrame({
    'Latitude': [30.483988, 29.846726, 29.938748, 30.031041, 32.797383,
                 30.065276, 30.526713, 30.240865, 29.933837,  29.995503, 30.476733, 32.470366],

    'Longitude': [-91.169601, -89.991966, -89.970822, -90.897768, -93.415004,
                  -90.593545, -91.746182, -93.272194, -89.944371, -90.409877, -91.210874, -93.789677],

    'Label': ['Baton Rouge Refinery', 'Alliance Refinery', 'Chalmette Refinery', 'Convent Refinery', 'Cotton Valley Refinery',
              'Garyville Refinery','Delek Refinery', 'Phillips 66 Lake Charles', 'Meraux Valero', 'Shell Norco',
              'Placid Refinery Port Allen', 'Calumet Shreveport']})

#convert lat/lon to Point geometry using shapely.point
geometry = [Point(xy) for xy in zip(points_df['Longitude'], points_df['Latitude'])]

#create another geodataframe out of these points
points_gdf = gpd.GeoDataFrame(points_df, geometry=geometry, crs='EPSG:4326')

incidence_plotter(merged_Louisiana, 'Louisiana', points_gdf)
```

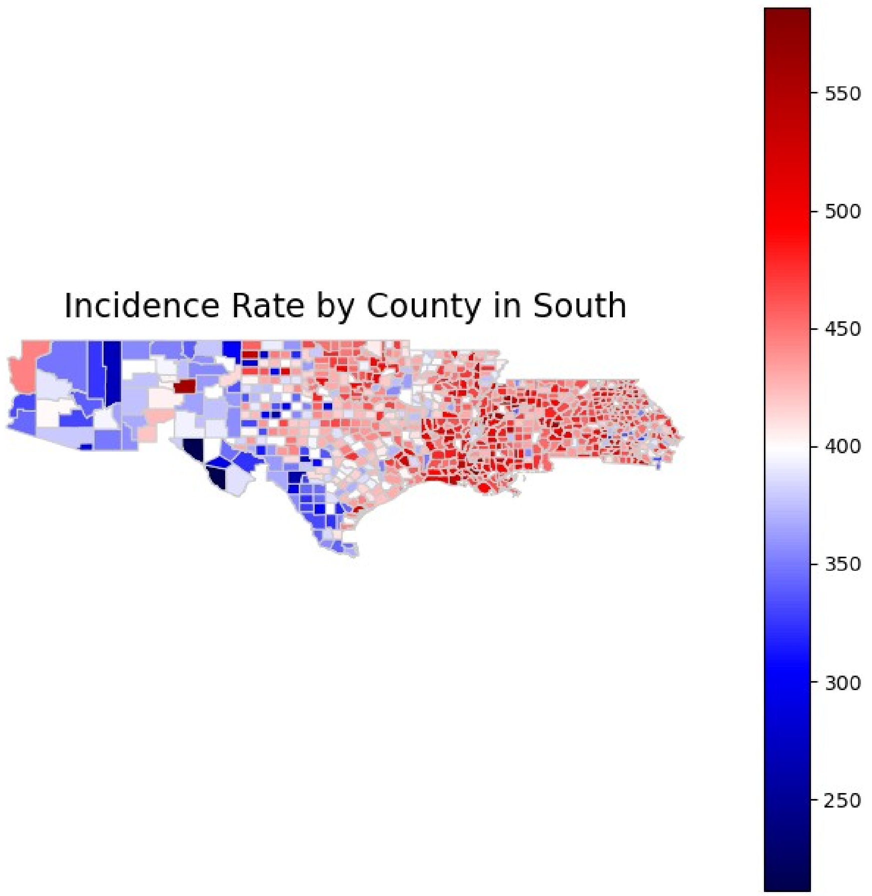
Python

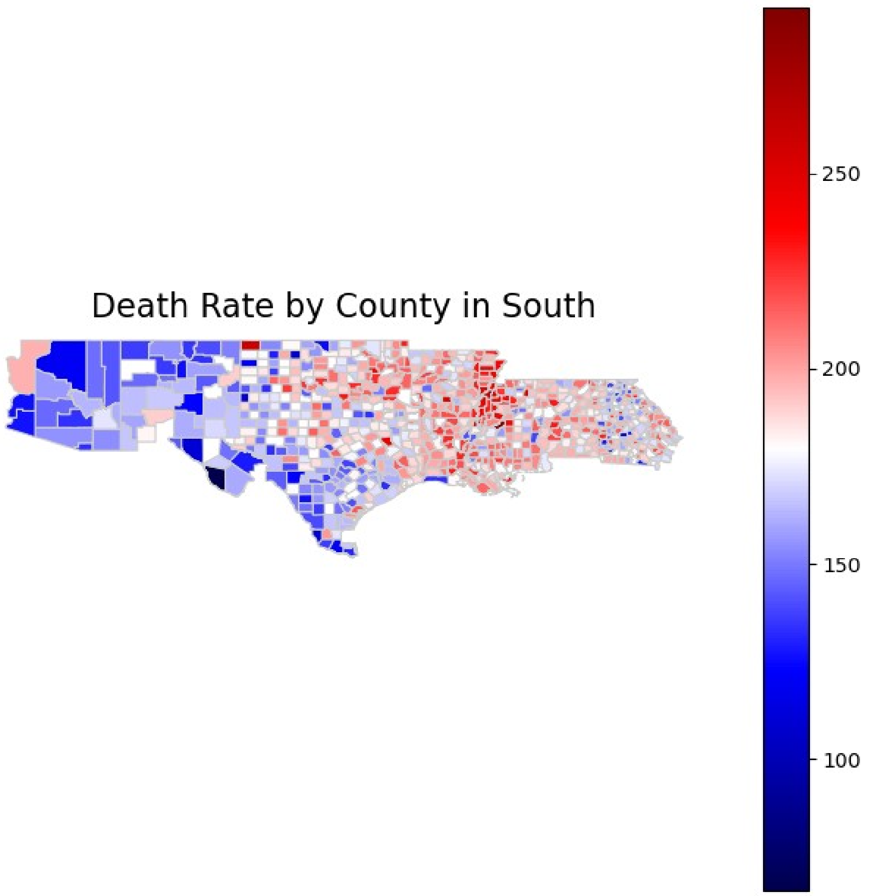
EDA - Expanded





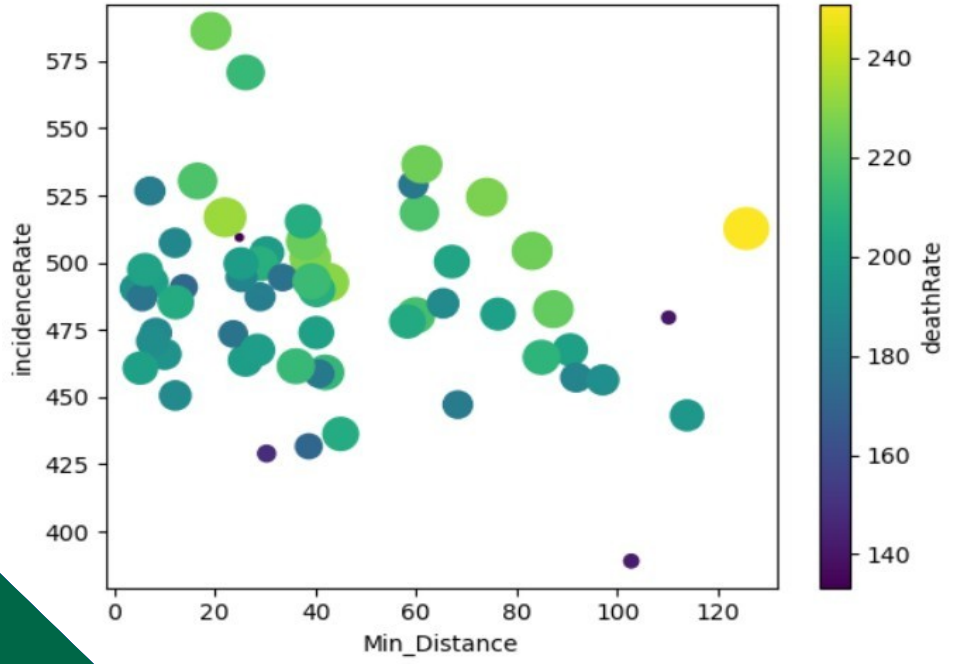
EDA - Expanded



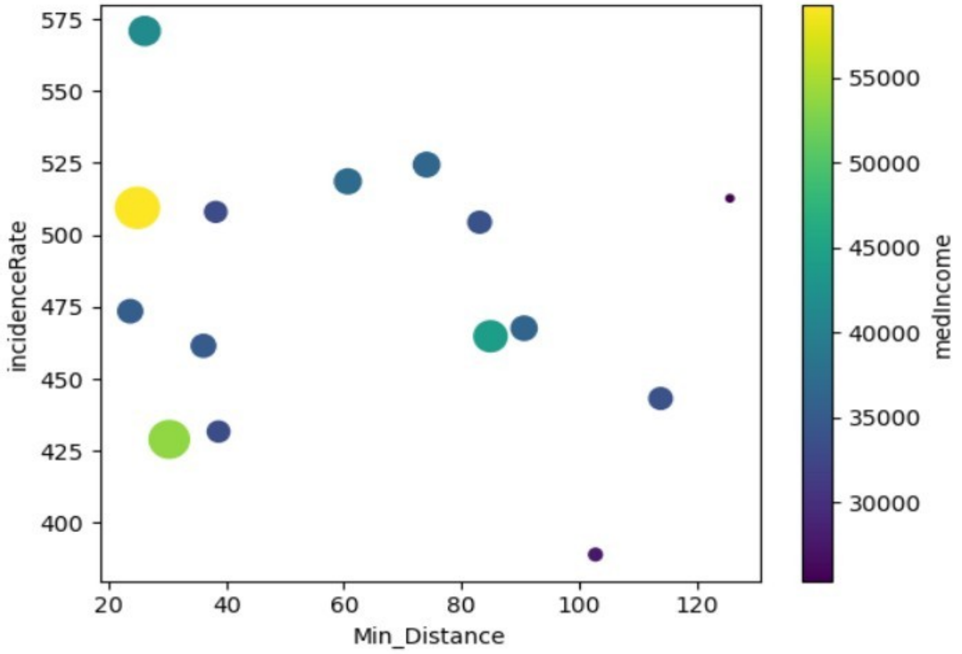


Correlation with Distance

Distance to Incidence Rate



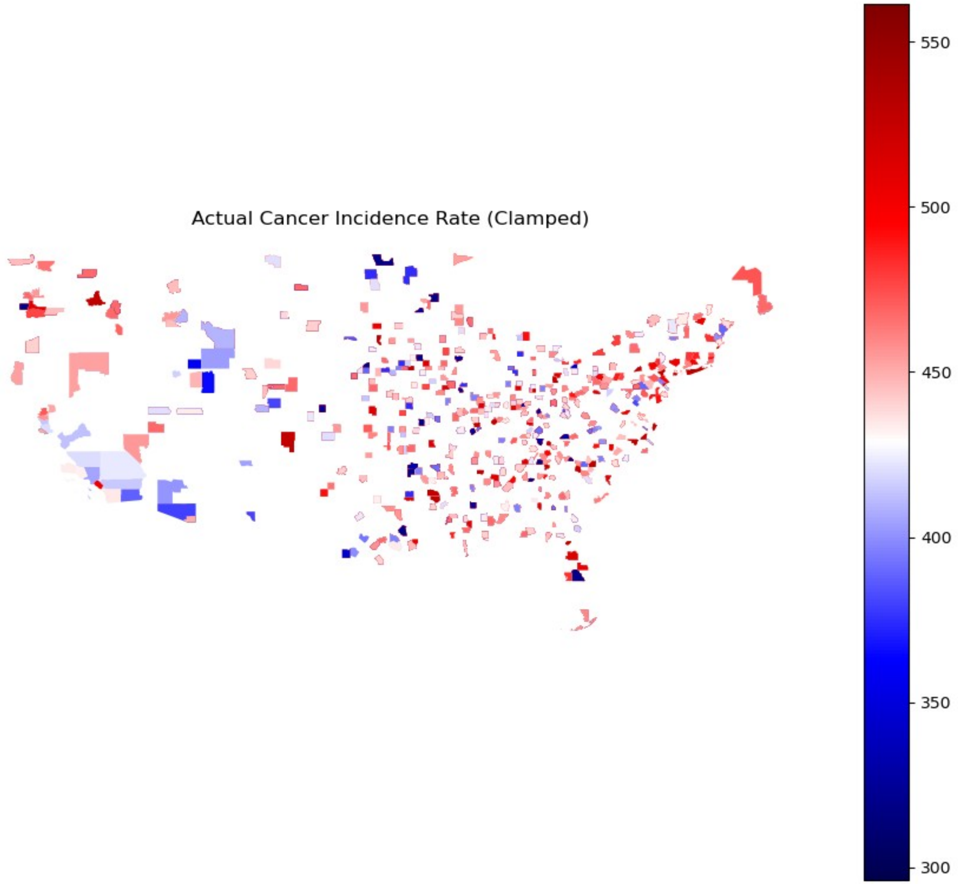
Distance to Incidence Rate:
Median Income below $20,000

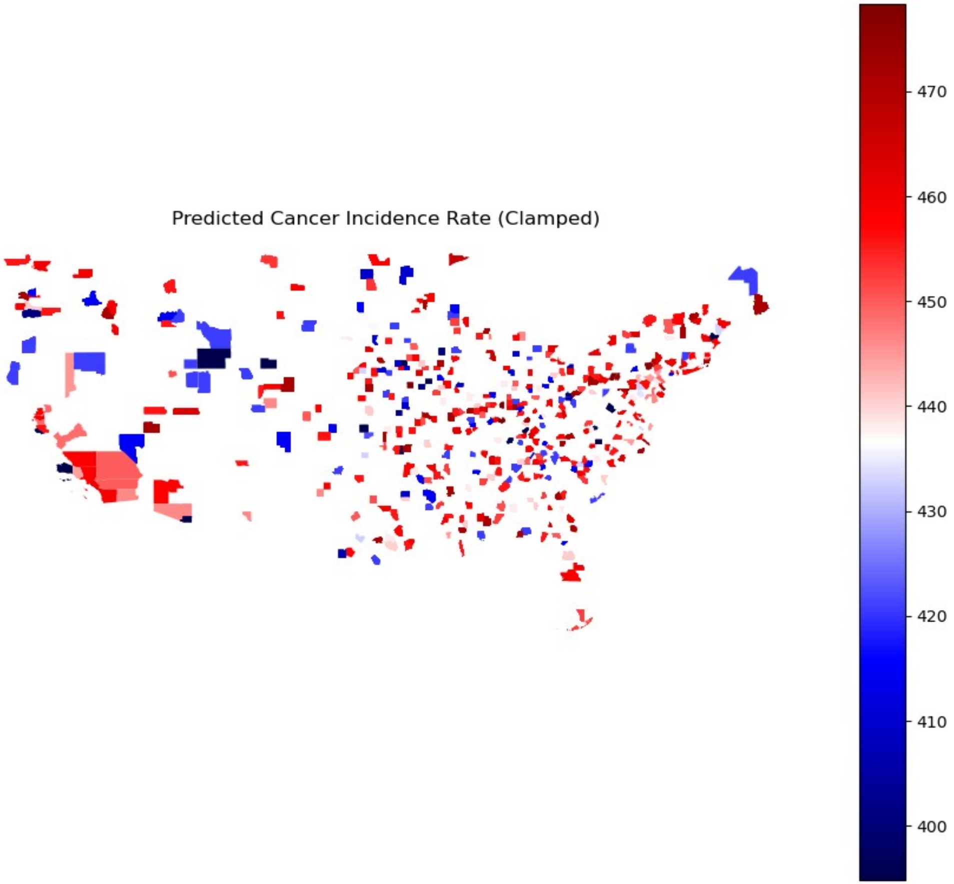


Model Training

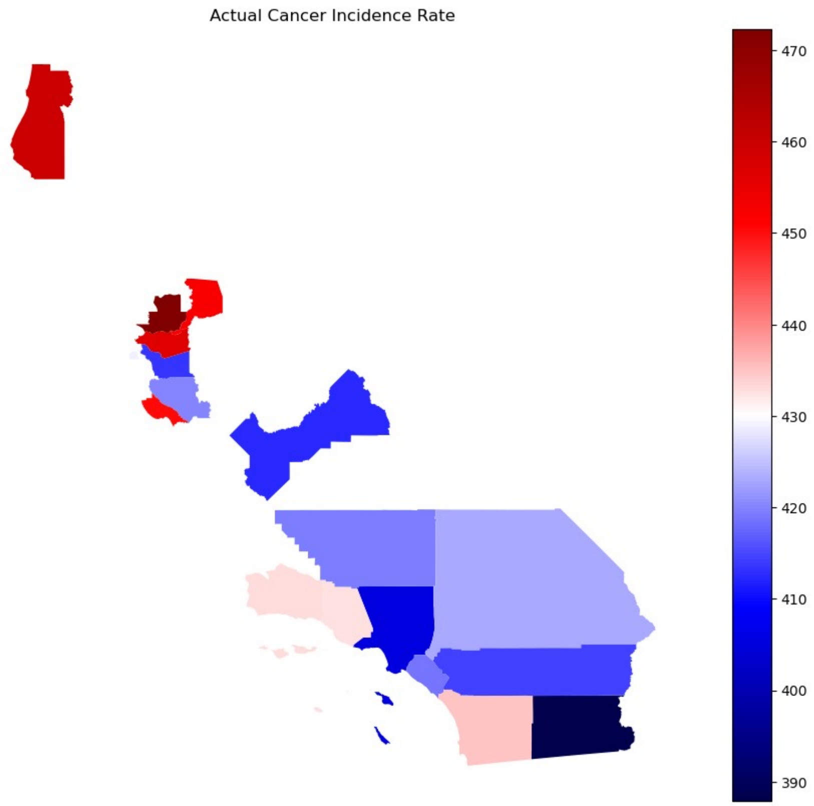
- A 10NN regressor model yielded decent results when trained on all states recorded pollution metrics.
- Train on pollution levels, predict cancer incidence.

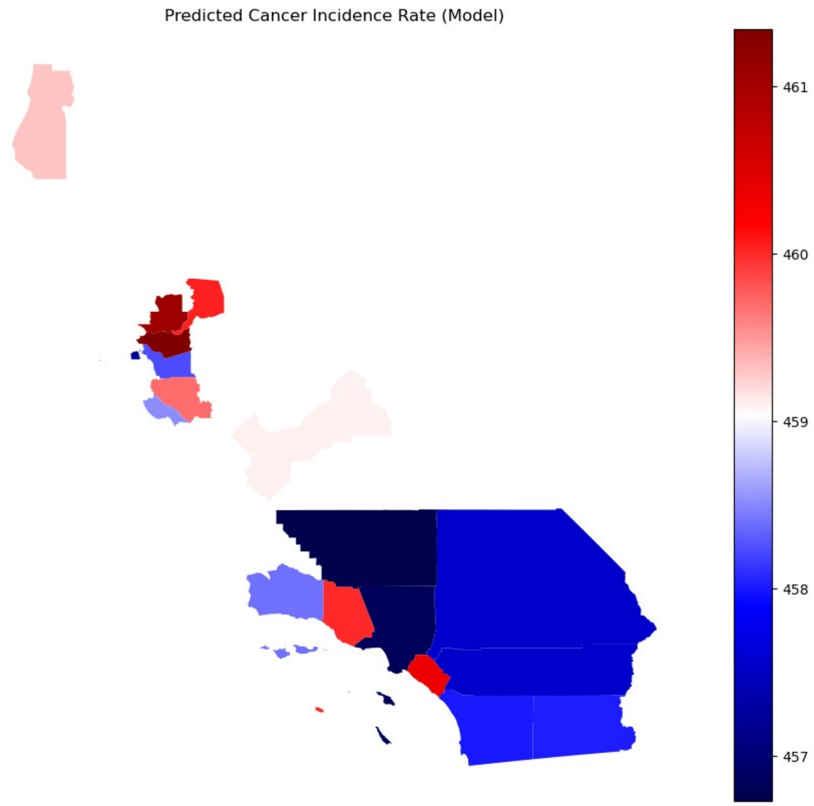
Minor Problem



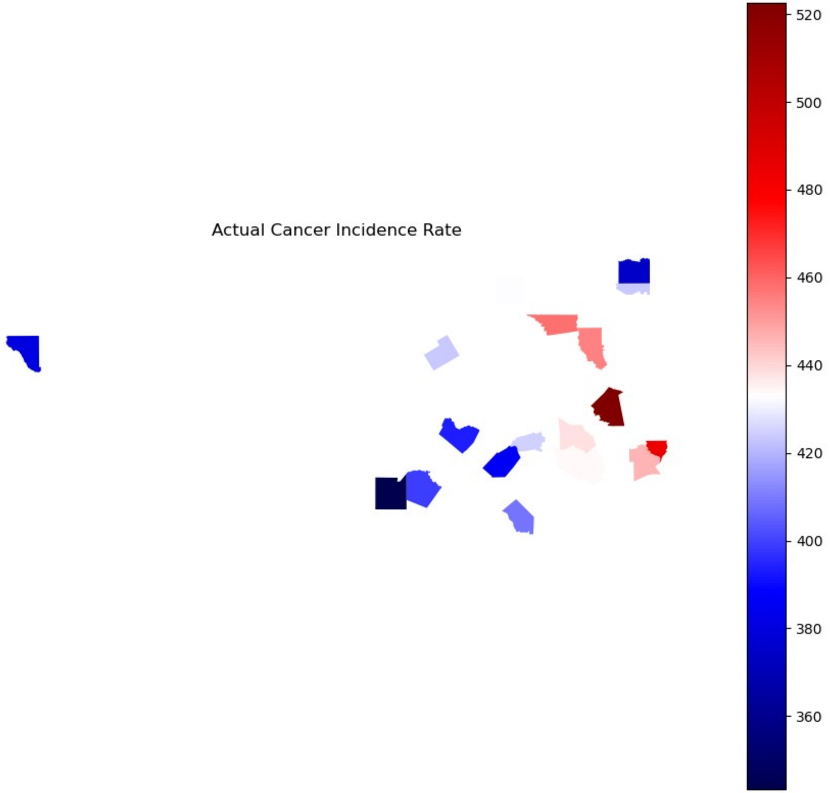


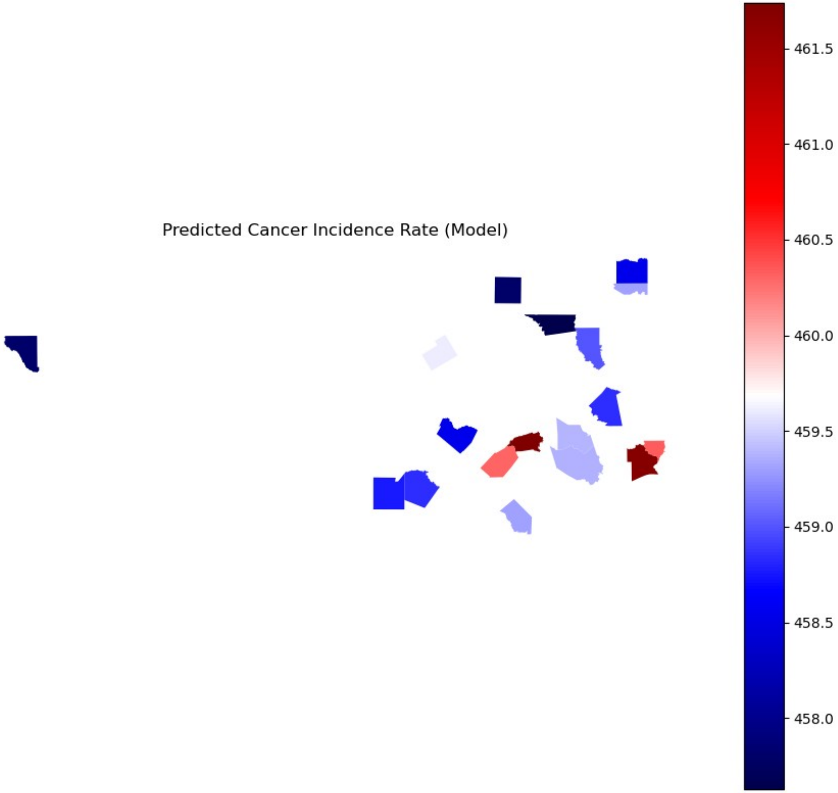
Model Results – California



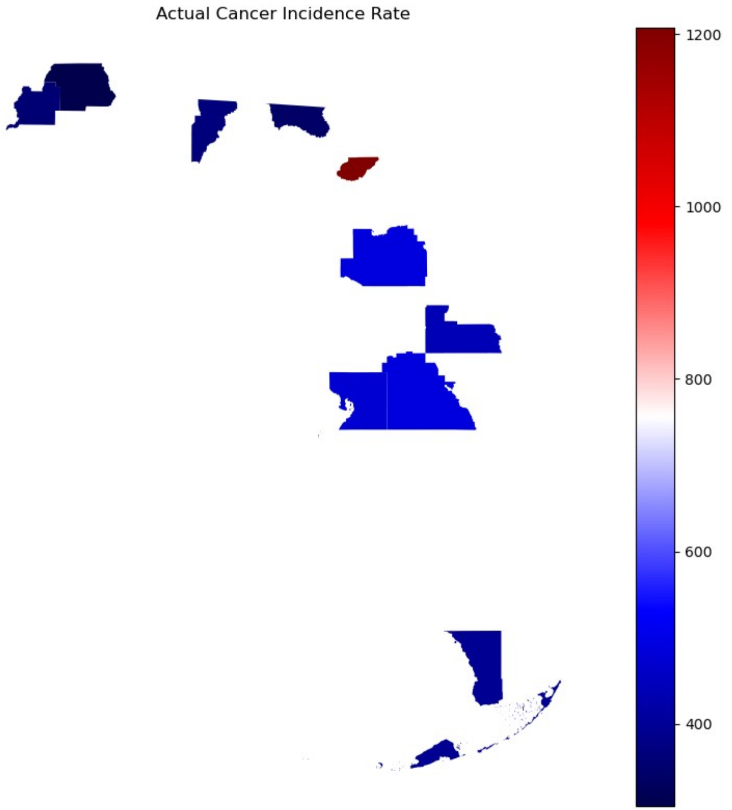


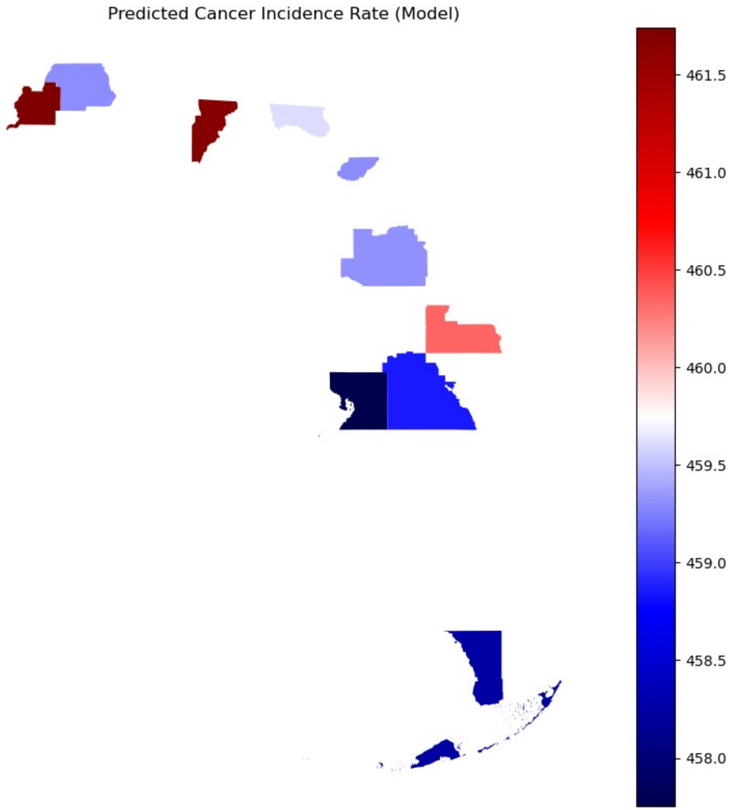
Model Results – Texas



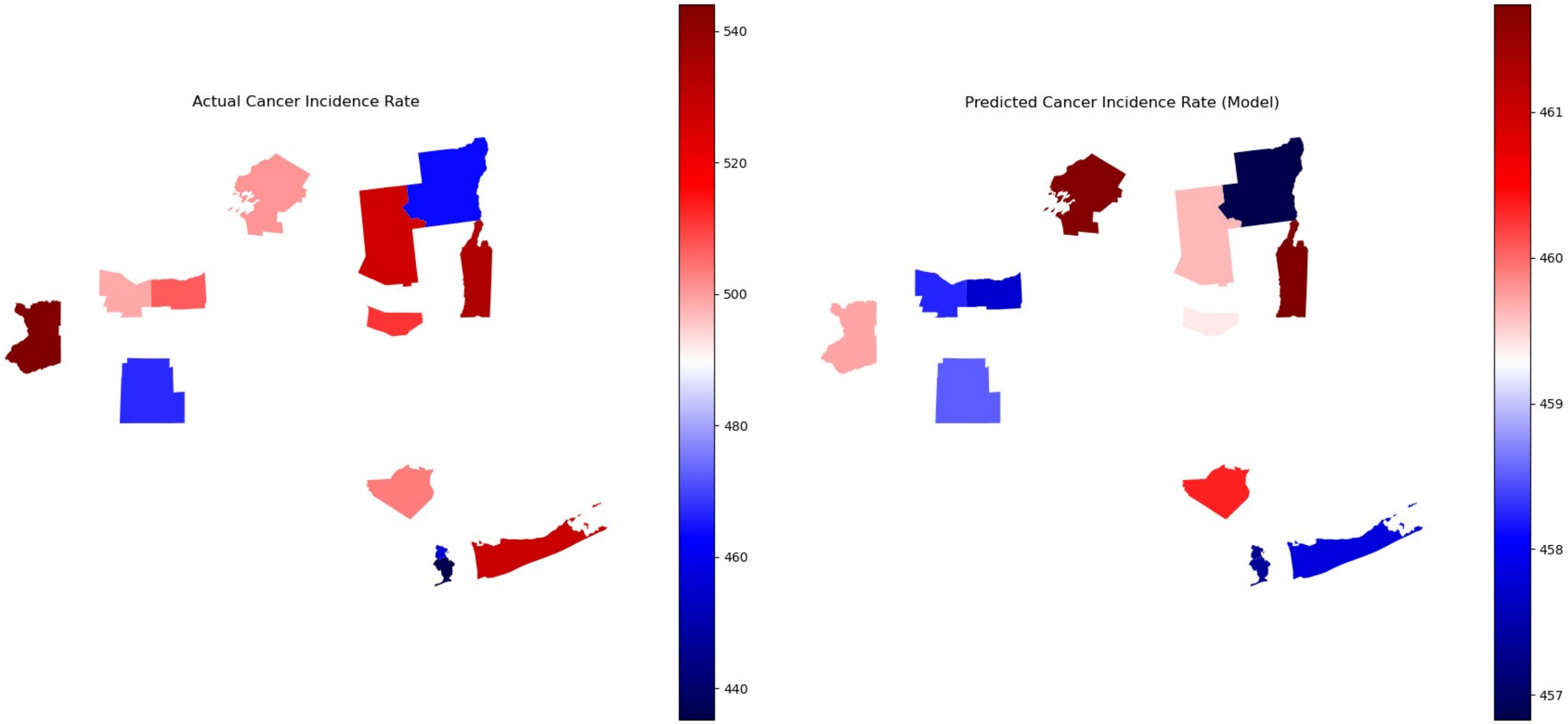


Model Results – Florida





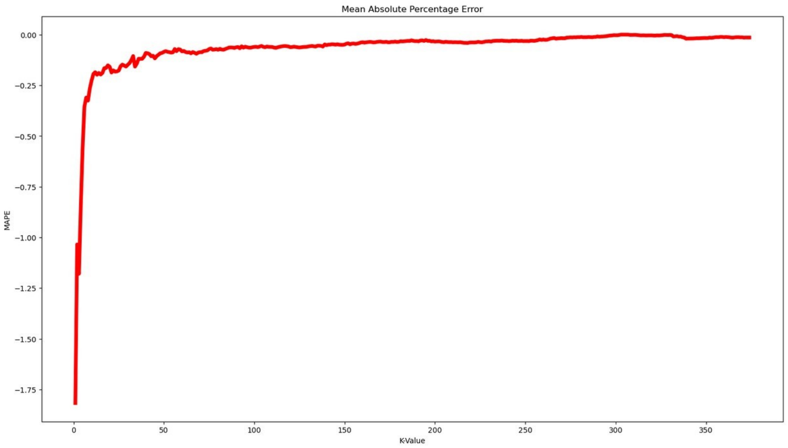
Model Results – New York



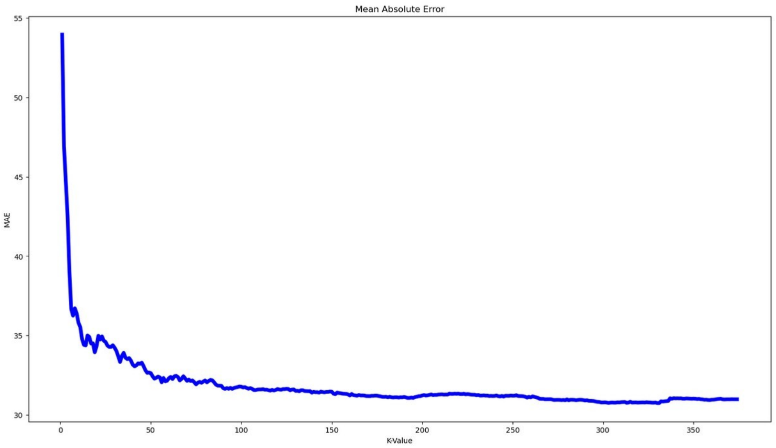
Parameter Tuning

- Need more data for such a complex relationship.
- Still see increasing performance.

R2 Score



Mean Absolute Error

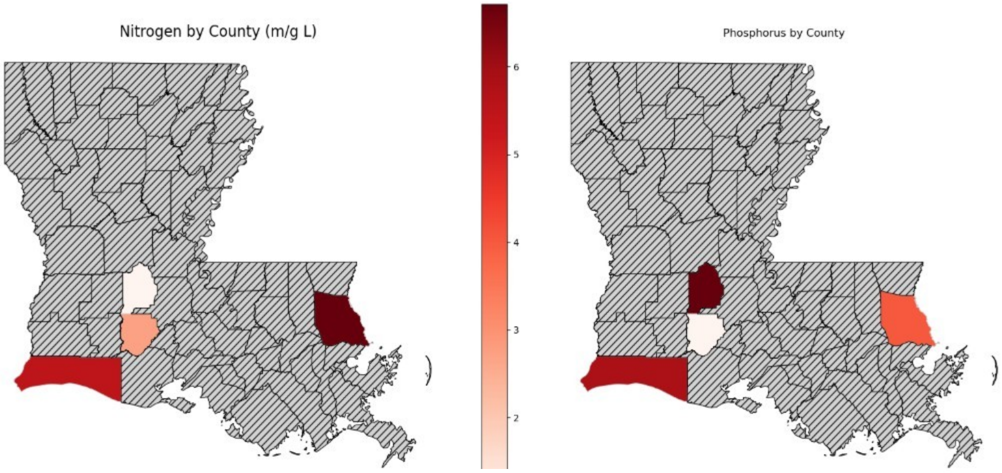


Model Results

- Single digit error rates on 80/20 train test split.
- Could see higher performance with more data, or more tuning.
- I would like to expand this dataset myself, with the distance to nearest plant metric as my pivot point.

Future Work

- Gather my own data
- Stop peeking



Takeaways

- Noticeable concentration near petrochemical plants and the Mississippi river
- Findings of the Tulane paper are upheld
- Insights gained from this data can be used to accurately predict trends elsewhere